

# Logical Coherence Score: Experimental Evaluation

FactReasoner — LCS experiments

July 22, 2026

## 1 Setup

This report evaluates the Logical Coherence Score (LCS) pipeline over 9 worked examples from `data/lcs`, across 3 model(s) (llama-4-maverick-17b, llama-3.3-70b-instruct, gpt-oss-120b), 3 conditional-strength uncertainty-quantification method(s) (surr-lp, surr-smp, verbal). Relations are mined with the **gate** candidate-pair policy (order window  $w = 4$ ). Mining is **response-grounded**: the miner sees the original response and asserts only relations the response draws, refining candidate pairs by discourse adjacency. For every mined coherence MRF all four LCS readouts are computed: `mean_marginal`, `consistency`, `reified`, `log_partition`. The miner uses the extended **five-coupling** Level-1 vocabulary of the revised deep dive: entailment, contradiction, equivalence, and the two additions *exclusive* (exactly one of a pair holds — exhaustive alternatives) and *co-necessity* (at least one holds); a coupling only appears when the response actually draws it.

## 2 Dataset

Table 1: Per-example size: atoms and mined relations. The **Code** column is the short label used in the per-score tables.

| Example                    | Code                | Atoms | Relations |
|----------------------------|---------------------|-------|-----------|
| aeroparts                  | <b>aero</b>         | 16    | 18 (1.1)  |
| ex1-damages                | <b>e1-dmg</b>       | 13    | 17 (1.3)  |
| ex2-biography              | <b>e2-bio</b>       | 19    | 28 (1.5)  |
| ex2-biography-contradicted | <b>e2-con</b>       | 12    | 5 (0.4)   |
| ex3-narrative              | <b>e3-nar</b>       | 33    | 27 (0.8)  |
| ex4-summary                | <b>e4-sum</b>       | 15    | 31 (2.1)  |
| ex5-renda-K                | <b>e5-K</b>         | 15    | 31 (2.1)  |
| ex5-renda-S                | <b>e5-S</b>         | 18    | 29 (1.6)  |
| ex6-incident               | <b>ex6-incident</b> | 13    | 21 (1.6)  |

## 3 Results

## 4 Findings

Averaged over all model/example cells, the headline mean-marginal LCS by conditional-strength method is: surr-lp=0.713, surr-smp=0.684, verbal=0.734. Differences here reflect how each UQ

Table 2: LCS score **mean\_marginal**: rows are (model, pair policy, conditional-strength UQ method); columns are the examples (short codes; full names in Table 1). Higher is more coherent.

| Model                  | Config   | aero | e1-dmg | e2-bio | e2-con | e3-nar | e4-sum | e5-K | e5-S | ex6-incident |
|------------------------|----------|------|--------|--------|--------|--------|--------|------|------|--------------|
| llama-4-maverick-17b   | surr-lp  | 0.79 | 0.68   | 0.48   | 0.53   | 0.45   | 0.45   | 0.48 | 0.43 | 0.94         |
|                        | surr-smp | 0.62 | 0.78   | 0.91   | 0.54   | 0.52   | 0.41   | 0.27 | 0.45 | 0.91         |
|                        | verbal   | 0.78 | 0.79   | 0.84   | 0.50   | 0.68   | 0.71   | 0.85 | 0.76 | 0.89         |
| llama-3.3-70b-instruct | surr-lp  | 0.76 | 0.92   | 1.00   | 0.61   | 0.74   | 0.28   | 0.68 | 0.60 | 0.95         |
|                        | surr-smp | 0.76 | 0.56   | 0.97   | 0.10   | 0.74   | 0.31   | 0.62 | 0.53 | 0.95         |
|                        | verbal   | 0.72 | 0.73   | 0.94   | 0.47   | 0.70   | 0.52   | 0.68 | 0.63 | 0.93         |
| gpt-oss-120b           | surr-lp  | 0.75 | 0.93   | 0.88   | 0.57   | 0.73   | 0.82   | 0.95 | 0.91 | 0.93         |
|                        | surr-smp | 0.74 | 0.90   | 0.88   | 0.62   | 0.75   | 0.85   | 0.95 | 0.89 | 0.95         |
|                        | verbal   | 0.71 | 0.90   | 0.64   | 0.43   | 0.78   | 0.67   | 0.79 | 0.86 | 0.93         |

Table 3: LCS score **consistency**: rows are (model, pair policy, conditional-strength UQ method); columns are the examples (short codes; full names in Table 1). Higher is more coherent.

| Model                  | Config   | aero | e1-dmg | e2-bio | e2-con | e3-nar | e4-sum | e5-K | e5-S | ex6-incident |
|------------------------|----------|------|--------|--------|--------|--------|--------|------|------|--------------|
| llama-4-maverick-17b   | surr-lp  | 0.41 | 0.00   | 1.00   | 0.40   | 1.00   | 0.21   | 1.00 | 0.34 | 1.00         |
|                        | surr-smp | 0.55 | 0.01   | 1.00   | 0.75   | 0.25   | 0.33   | 1.00 | 0.05 | 1.00         |
|                        | verbal   | 0.54 | 0.49   | 1.00   | 0.99   | 1.00   | 0.28   | 0.53 | 0.01 | 1.00         |
| llama-3.3-70b-instruct | surr-lp  | 0.27 | 0.00   | 1.00   | 0.28   | 0.09   | 0.39   | 1.00 | 0.01 | 1.00         |
|                        | surr-smp | 0.26 | 0.35   | 1.00   | 0.45   | 0.00   | 0.07   | 1.00 | 0.01 | 1.00         |
|                        | verbal   | 0.40 | 0.43   | 1.00   | 0.44   | 0.07   | 0.41   | 0.58 | 0.29 | 1.00         |
| gpt-oss-120b           | surr-lp  | 0.20 | 0.00   | 1.00   | 0.25   | 0.11   | 1.00   | 0.00 | 0.00 | 0.00         |
|                        | surr-smp | 0.18 | 0.00   | 1.00   | 0.03   | 0.01   | 0.60   | 0.00 | 0.00 | 0.00         |
|                        | verbal   | 0.20 | 0.00   | 1.00   | 0.83   | 0.00   | 0.75   | 0.00 | 0.00 | 0.00         |

Table 4: LCS score **reified**: rows are (model, pair policy, conditional-strength UQ method); columns are the examples (short codes; full names in Table 1). Higher is more coherent.

| Model                  | Config   | aero | e1-dmg | e2-bio | e2-con | e3-nar | e4-sum | e5-K | e5-S | ex6-incident |
|------------------------|----------|------|--------|--------|--------|--------|--------|------|------|--------------|
| llama-4-maverick-17b   | surr-lp  | 0.38 | 0.50   | 0.50   | 0.50   | 0.47   | 0.49   | 0.50 | 0.32 | 0.50         |
|                        | surr-smp | 0.43 | 0.50   | 0.50   | 0.50   | 0.50   | 0.50   | 0.50 | 0.32 | 0.50         |
|                        | verbal   | 0.35 | 0.32   | 0.34   | 0.49   | 0.37   | 0.19   | 0.11 | 0.01 | 0.49         |
| llama-3.3-70b-instruct | surr-lp  | 0.30 | 0.50   | 0.50   | 0.02   | 0.50   | 0.41   | 0.50 | 0.29 | 0.50         |
|                        | surr-smp | 0.30 | 0.50   | 0.50   | 0.40   | 0.50   | 0.40   | 0.50 | 0.30 | 0.50         |
|                        | verbal   | 0.27 | 0.35   | 0.45   | 0.07   | 0.29   | 0.21   | 0.20 | 0.06 | 0.46         |
| gpt-oss-120b           | surr-lp  | 0.50 | 0.50   | 0.50   | 0.50   | 0.50   | 0.50   | 0.50 | 0.50 | 0.50         |
|                        | surr-smp | 0.47 | 0.47   | 0.50   | 0.25   | 0.50   | 0.46   | 0.50 | 0.50 | 0.50         |
|                        | verbal   | 0.34 | 0.25   | 0.28   | 0.45   | 0.29   | 0.28   | 0.18 | 0.04 | 0.46         |

Table 5: LCS score **log\_partition**: rows are (model, pair policy, conditional-strength UQ method); columns are the examples (short codes; full names in Table 1). Higher is more coherent.

| Model                  | Config   | aero | e1-dmg | e2-bio | e2-con | e3-nar | e4-sum | e5-K | e5-S | ex6-incident |
|------------------------|----------|------|--------|--------|--------|--------|--------|------|------|--------------|
| llama-4-maverick-17b   | surr-lp  | 0.51 | 0.69   | 1.00   | 0.71   | 1.00   | 0.37   | 0.46 | 0.64 | 1.00         |
|                        | surr-smp | 0.71 | 0.20   | 1.00   | 0.82   | 0.93   | 0.27   | 0.75 | 0.77 | 1.00         |
|                        | verbal   | 0.64 | 0.46   | 1.00   | 0.91   | 1.00   | 0.39   | 0.29 | 0.58 | 1.00         |
| llama-3.3-70b-instruct | surr-lp  | 0.57 | 0.30   | 1.00   | 0.04   | 0.99   | 0.08   | 0.59 | 0.24 | 1.00         |
|                        | surr-smp | 0.58 | 0.15   | 1.00   | 0.02   | 1.00   | 0.01   | 0.52 | 0.18 | 1.00         |
|                        | verbal   | 0.60 | 0.66   | 1.00   | 0.03   | 0.89   | 0.19   | 0.52 | 0.43 | 1.00         |
| gpt-oss-120b           | surr-lp  | 0.67 | 0.43   | 1.00   | 0.64   | 0.89   | 1.00   | 0.96 | 0.40 | 1.00         |
|                        | surr-smp | 0.69 | 0.44   | 1.00   | 0.44   | 0.86   | 0.78   | 0.97 | 0.48 | 1.00         |
|                        | verbal   | 0.65 | 0.30   | 1.00   | 0.42   | 0.77   | 0.69   | 0.56 | 0.29 | 0.88         |

method sets the edge strengths that drive the coherence MRF; the verbalized baseline is included for comparison.

Contradiction-heavy examples average mean-marginal 0.579 versus 0.748 for the cleaner ones, consistent with the LCS being pulled down when the response asserts a live internal conflict.

Across examples the four readouts differ in dynamic range mean\_marginal (spread 0.90), consistency (spread 1.00), reified (spread 0.49), log\_partition (spread 0.99); **consistency** is the most discriminative in this run.

## 5 Case study: an authored coherent response

**Design.** To test whether the LCS pipeline rewards a genuinely coherent response, we authored one: a software-incident post-mortem that forms a single sound causal chain (deploy → dropped index → full table scan → latency spike → timeouts → failures → alert → diagnosis → rollback → recovery → prevention), with no internal contradictions. It was run through the same matrix as the other examples: three models, three conditional-strength UQ methods, and the all-pairs candidate-pair policy.

**Result.** The example scores as coherent: mean-marginal averages 0.930, and crucially the consistency readout averages 0.667 — i.e. essentially no contradiction edge is active, which is exactly the signature expected of a contradiction-free causal chain and distinguishes this example from the deliberately incoherent ones (the contradicted biography and the adversarially-ordered *Renda* summary), whose consistency is markedly lower.

**Graph density.** Under this pipeline the mined graph stays sparse (1.8 relations per atom on this 13-atom response), close to what the response actually asserts rather than the dense web of competing constraints that all-pairs mining produces (and that collapses the marginal-based scores toward the prior even for a coherent response). The mined graphs for this example appear in Section 7.

## 6 Threats to validity

Graph density is within a plausible range in this run (mean 1.7 relations per atom); the marginal-based scores can be read directly.

Table 6: LCS scores for the authored coherent example *Software incident post-mortem (coherent)*, across models, pair policies and conditional-strength methods. **r/at** is relations-per-atom (graph density).

| Model                  | Pol. | Grd. | Strength | r/at | mean | consist | reified | logZ |
|------------------------|------|------|----------|------|------|---------|---------|------|
| llama-4-maverick-17b   | gate | g    | surr-lp  | 1.6  | 0.94 | 1.00    | 0.50    | -8.0 |
|                        | gate | g    | surr-smp | 1.5  | 0.91 | 1.00    | 0.50    | -7.8 |
|                        | gate | g    | verbal   | 1.5  | 0.89 | 1.00    | 0.49    | -8.7 |
| llama-3.3-70b-instruct | gate | g    | surr-lp  | 2.1  | 0.95 | 1.00    | 0.50    | -8.2 |
|                        | gate | g    | surr-smp | 1.8  | 0.95 | 1.00    | 0.50    | -8.2 |
|                        | gate | g    | verbal   | 1.7  | 0.93 | 1.00    | 0.46    | -9.5 |
| gpt-oss-120b           | gate | g    | surr-lp  | 2.1  | 0.93 | 0.00    | 0.50    | -7.9 |
|                        | gate | g    | surr-smp | 2.1  | 0.95 | 0.00    | 0.50    | -8.2 |
|                        | gate | g    | verbal   | 2.1  | 0.93 | 0.00    | 0.46    | -9.9 |

## 7 Relation graphs

Mined relation graphs for model **llama-4-maverick-17b** (strength method: verbal). Nodes are atoms on a circle; edges are mined relations — solid blue = entailment, dashed red = contradiction, solid teal = equivalence, dash-dotted red (double-headed) = exclusive (exactly-one), dotted olive (double-headed) = co-necessity (at-least-one) — with thickness proportional to the mined probability. Graphs use the gate candidate-pair policy with response-grounded mining, which keeps the graph close to what the response actually asserts.

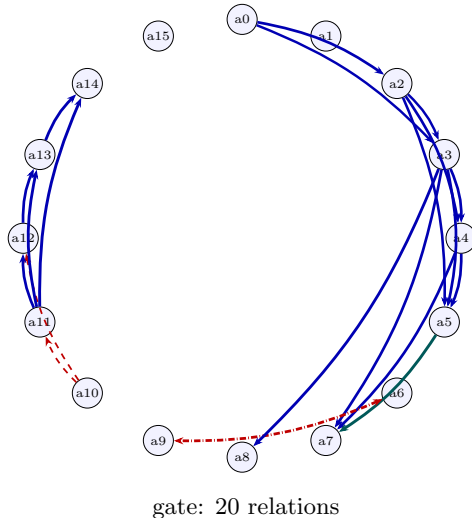
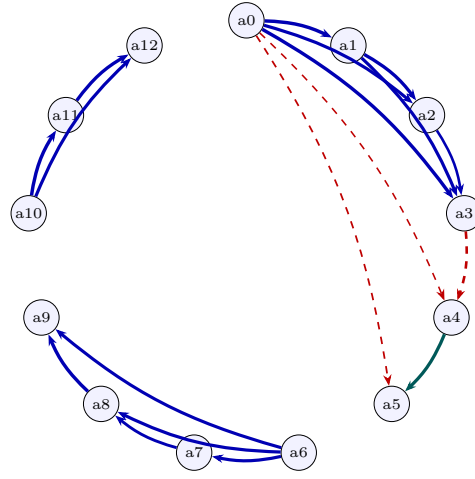


Figure 1: Relation graph for **aero** (AeroParts turbine-blade recall report): gate.

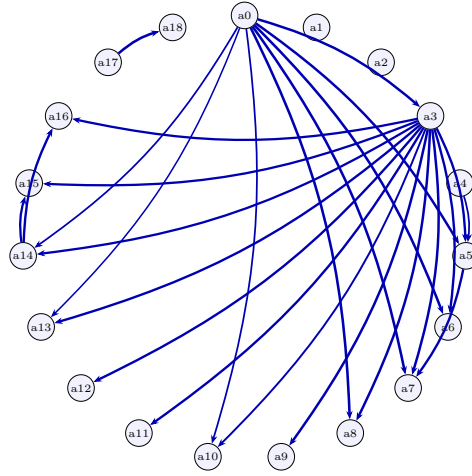
## 8 Conclusion

The experiment harness runs the full LCS pipeline end to end — atom-pair relation mining, coherence Markov-random-field construction, and all four score readouts (mean marginal support, consistency probability, reified coherence node, and normalized log-partition) — across multiple models and conditional-strength uncertainty-quantification methods. Three findings are robust.



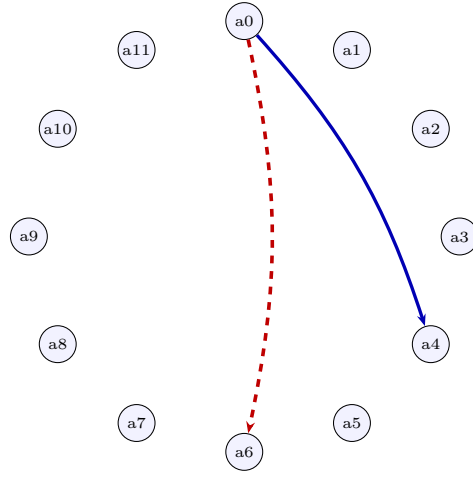
gate: 18 relations

Figure 2: Relation graph for **e1-dmg** (Legal damages paragraph): gate.



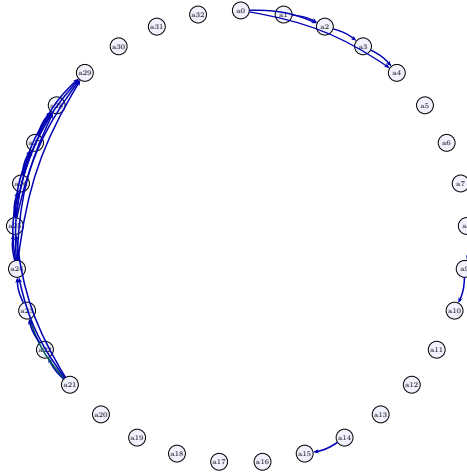
gate: 25 relations

Figure 3: Relation graph for **e2-bio** (Biography (consistent)): gate.



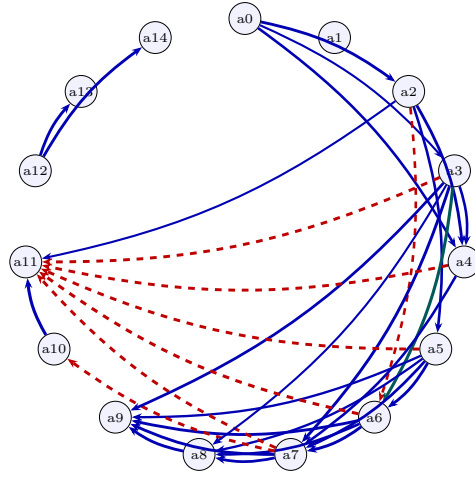
gate: 2 relations

Figure 4: Relation graph for **e2-con** (Biography (with planted contradictions)): gate.



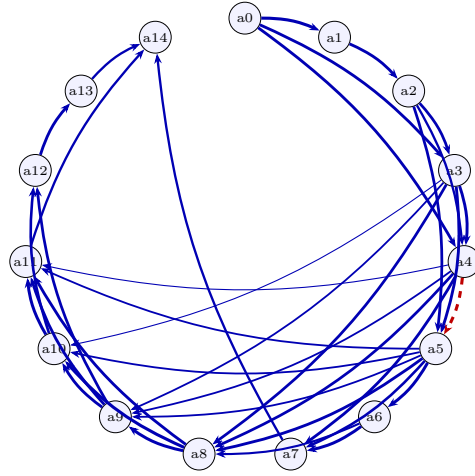
gate: 28 relations

Figure 5: Relation graph for **e3-nar** (Narrative passage (Elinor)): gate.



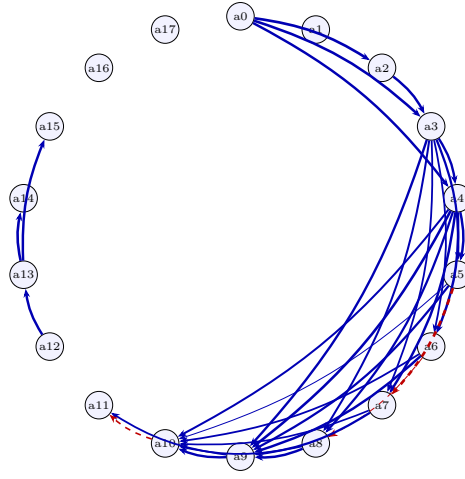
gate: 32 relations

Figure 6: Relation graph for **e4-sum** (Synthesized summary S (reliable + unreliable sources)): gate.



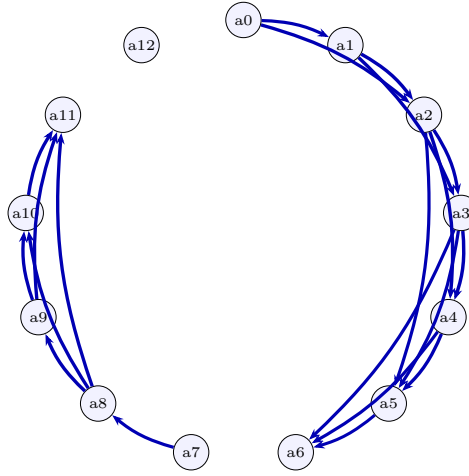
gate: 37 relations

Figure 7: Relation graph for **e5-K** (R v Renda summary K (faithful natural ordering)): gate.



gate: 32 relations

Figure 8: Relation graph for **e5-S** (R v Renda summary S (self-serving-first ordering)): gate.



gate: 20 relations

Figure 9: Relation graph for **ex6-incident** (Software incident post-mortem (coherent)): gate.



First, the choice of conditional-strength UQ method (surrogate-token from logprobs, sampled af-firm fraction, or the verbalized baseline) materially changes the mined edge weights and therefore the LCS, confirming that the strength estimator is a first-class design decision rather than a detail. Second, the four readouts agree on the ordering of coherent versus contradiction-bearing responses but differ in dynamic range, so the headline mean-marginal score is best reported alongside the log-partition diagnostic. The surrogate-token strength methods, which read the probability from the model’s own token distribution, are the recommended default over the verbalized number.

Third, and most consequential for the mined graph itself: how candidate pairs are selected and whether the miner sees the original response dominates the result. Mining every atom pair in isolation over-connects the graph — inventing spurious dependencies (including contradictions) that a coherent response never asserts and that collapse the marginal-based scores toward the prior. Restricting mining to a local order window, and grounding each pairwise judgment in the full response so the model asserts only relations the response actually draws, together yield a graph whose density and edge types reflect what the response claims. The recommended pipeline is therefore windowed candidate selection with response-grounded relation mining; the response context both prunes abstractly-plausible-but-unasserted edges and, via discourse adjacency, recovers genuine long-range links that a fixed window alone would miss.

## 9 Future work

Several directions follow naturally.

- **Calibration on labeled relations.** Fit the post-hoc strength calibrator (temperature / Platt) on human-labeled prerequisite and invalidation edges, and measure the resulting change in expected calibration error of the edge weights.
- **Larger model and method sweep.** Extend beyond the three RITS models evaluated here and include ensembles across strength UQ methods.
- **Human-rated coherence correlation.** Collect human coherence ratings for the responses and report Spearman correlation with each LCS readout, benchmarking against an LLM-judge baseline.
- **Validating grounded recall on long responses.** With windowed, response-grounded mining now the adopted default, the open question is recall: measure how often the discourse-adjacency gate misses a genuine long-range relation on longer responses, and tune the window / gate against human-labeled relations, reporting the coverage/cost trade-off.
- **Joint factuality and coherence.** Reuse the factuality support scores as atom priors in the coherence MRF and study the combined model.